Local Income Taxes and Residential Sorting:

Evidence from LEHD Origin-Destination Data

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Abstract

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1 Introduction

2 Institutional Background

2.1 Ohio Municipal Income Tax

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3 Data

3.1 LODES Origin-Destination Data

3.2 Municipal Tax Rate Panels

3.3 Municipal Financial Reports

3.4 Outcome Variable: share_live_in

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
Panel A: Pennsylvania (2002–2019)					
share_live_in	0.149	0.127	0.097	0	0.700
Resident EIT rate (%)	0.542	0.500	0.199	0	4.331
Total jobs	3,624	1,224			
Resident workers	674	154			
Panel B: Ohio (2002–2019)					
share_live_in	0.119	0.099	0.085		
Municipal tax rate (%)	1.222	1.500	0.821	0	3.000
Panel dimensions					
PA municipality × years	24,528 (1,383 municipalities)				
OH municipality × years	7,718 (430 municipalities)				

Notes: Municipality-year panel, 2002–2019. Sample restricted to municipalities with > 50 total jobs. share_live_in = workers who live and work in municipality / total workers in municipality. OH municipalities are Census places; PA municipalities are county subdivisions.

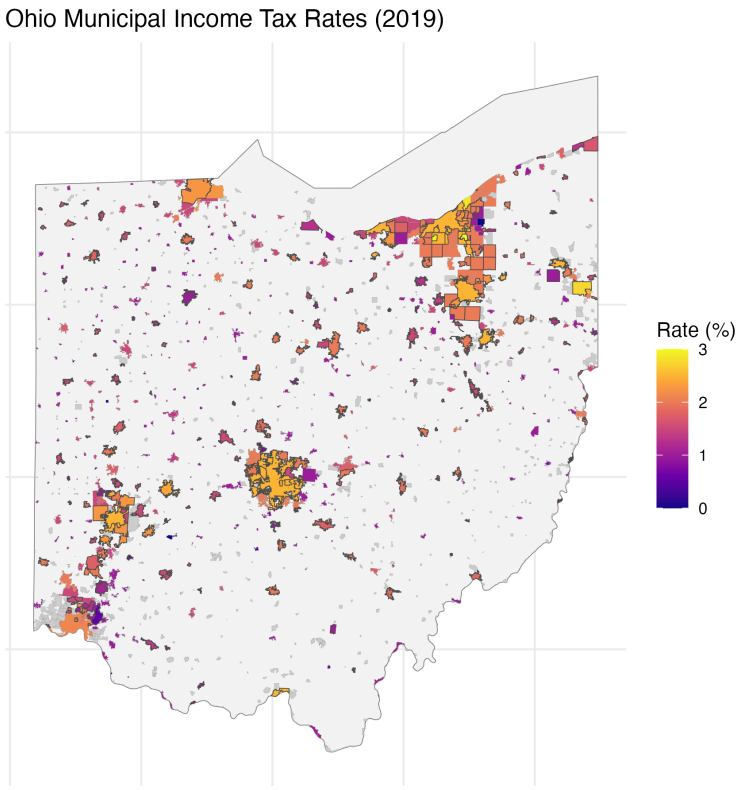


Figure 1: Ohio Municipal Income Tax Rates (2019). Bold outline = municipalities that raised rates during the sample period. Grey fill = municipalities with no municipal income tax.

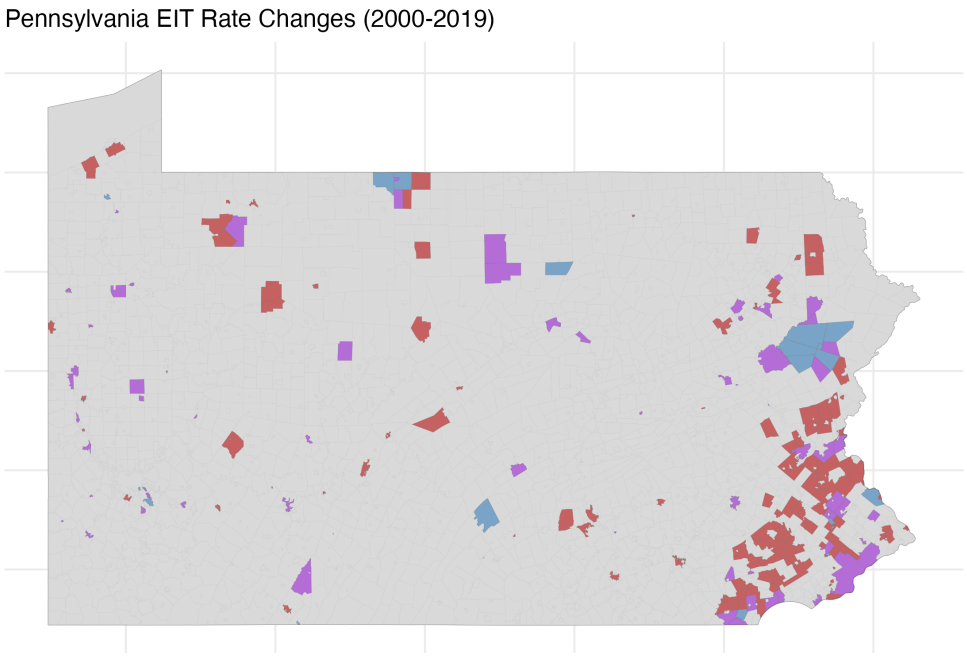


Figure 2: Pennsylvania EIT Rate Changes (2000–2019). Red = increased; blue = decreased; purple = both; grey = no change.

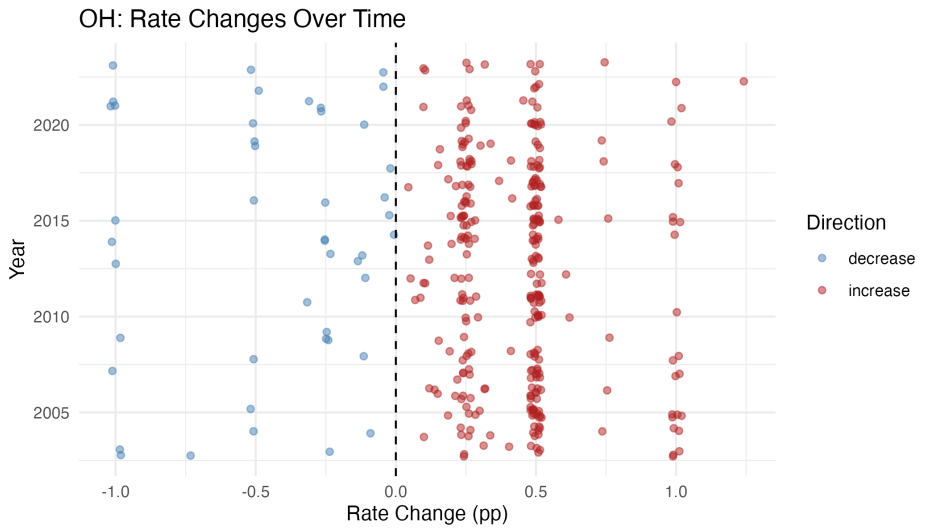


Figure 3: Ohio: Distribution of Municipal Income Tax Rate Changes Over Time

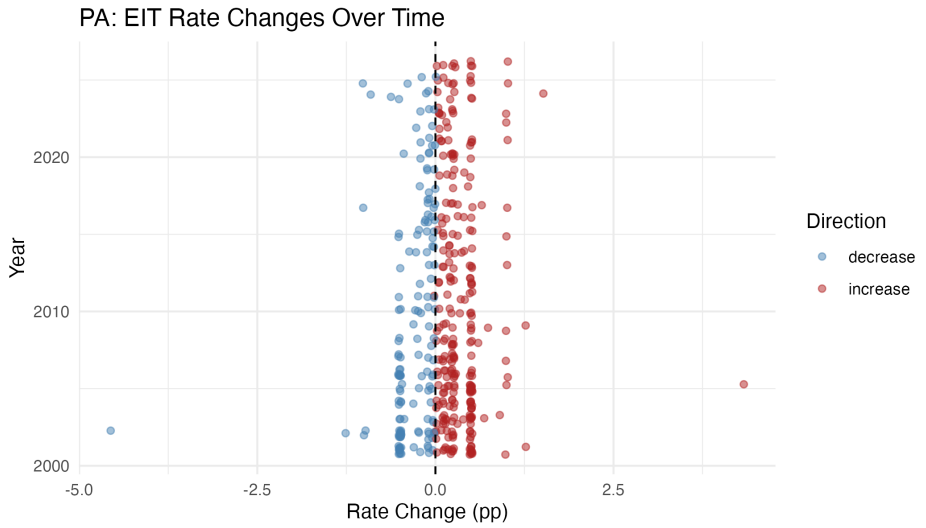


Figure 4: Pennsylvania: Distribution of EIT Rate Changes Over Time

4 Empirical Strategy

We organize our analysis around four questions, each building on the previous. All specifications use municipality and year fixed effects with standard errors clustered at the municipality level. For binary treatments under staggered timing, we use the Sun and Abraham (2021) interaction-weighted estimator:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \cdot D_{it}^{\ell} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{1}$$

where Y_{it} is share_live_in, D_{it}^{ℓ} indicates municipality i being ℓ years from its first treatment, and $\mathbf{X}_{it}$ is a vector of fiscal controls (available for PA). Pre-treatment coefficients ($\ell < -1$) test parallel trends. For continuous treatments (the actual tax rate), we use the de Chaisemartin and D'Haultfoeuille (2024) estimator, which handles both staggered timing and varying treatment intensity.

Q1: Does adopting a local income tax cause residential sorting? Ohio municipalities can adopt a local income tax; unincorporated townships have no tax. We compare municipalities that adopted a tax during the panel (going from 0 to a positive rate) against

no-tax townships. This is the cleanest Tiebout test: a discrete switch from no tax to having one. Only Ohio provides this extensive-margin variation; all Pennsylvania municipalities levy an EIT throughout the panel.

Q2: Do rate increases cause additional sorting? Conditional on already having a tax, does raising the rate push additional residents out? We estimate Equation 1 with rate increasers as the treated group. For Ohio, we use no-tax townships as the control group (stable-rate taxed cities produce null results, likely because they share unobserved urban characteristics with treated cities). For Pennsylvania, the control group is the 2,209 municipalities that never changed their EIT rate. We also estimate the DCDH continuous treatment model, which tests whether rate *magnitude* predicts sorting beyond the binary treatment effect.

Q3: Do rate decreases reverse sorting? We estimate Equation 1 with rate decreasers as the treated group and never-changed municipalities as controls. If sorting is reversible, decreases should increase the locally-resident share.

Q4: Is the sorting response asymmetric? We test whether increases and decreases have symmetric effects using a joint model with both indicators and a split cumulative dose specification:

$$Y_{it} = \alpha_i + \gamma_t + \beta^+ \cdot R_{it}^+ + \beta^- \cdot R_{it}^- + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{2}$$

where R_{it}^+ and R_{it}^- are cumulative rate increases and decreases from baseline. If $|\beta^+| > |\beta^-|$, residential location decisions are sticky: tax increases push residents out, but tax decreases do not pull them back.

5 Results

5.1 Q1: Does Adopting a Tax Cause Sorting?

We pool municipalities from both states that adopted a local income tax during the panel (going from a zero rate to a positive rate) and compare them to municipalities that never levied a tax. Only 8 municipalities adopted a tax and appear in the LODES data with sufficient employment: 1 in Ohio and 7 in Pennsylvania. The control group consists of 109 never-taxed municipalities (104 Ohio townships, 5 Pennsylvania municipalities). We include state × year fixed effects to absorb cross-state differences.

Table 2: Q1: Effect of Tax Adoption on share_live_in

	Pooled (OH + PA)
Sun-Abraham ATT	−0.0036
	(0.0066)
Treated	8 adopters (1 OH, 7 PA)
Control	109 never-taxed (104 OH, 5 PA)
Fixed Effects	Municipality, State × Year
Clusters	Municipality
Min. total jobs	10

Notes: Sun-Abraham ATT from pooled OH and PA sample. Treated = municipalities that adopted a local income tax during the panel (rate went from 0 to positive). Control = municipalities that never levied a tax in either state. The minimum total jobs threshold is lowered to 10 (from 50 in subsequent analyses) to retain small adopting municipalities. Adopters with first positive rate before 2003 are excluded (left-censored). The small treatment group (8 municipalities) limits statistical power. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

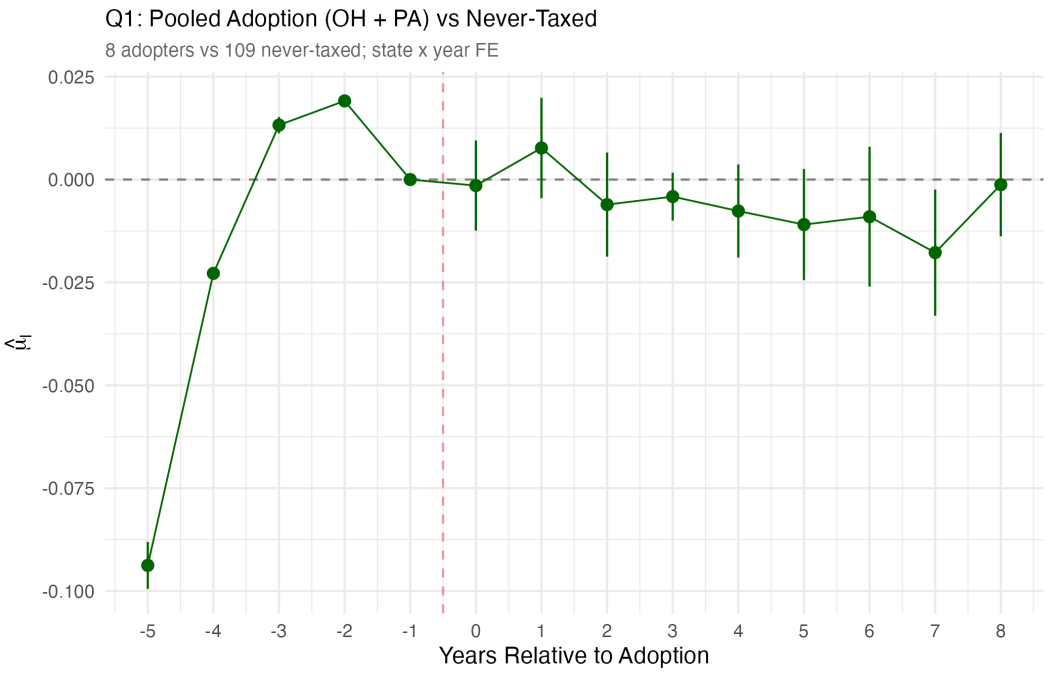


Figure 5: Q1: Pooled Adoption Event Study (OH + PA). 8 adopters vs 109 never-taxed controls. State × year FE. Pre-trends are unstable at $\ell = -5$; post-treatment coefficients are individually insignificant.

A1: We cannot reject the null. The pooled ATT is −0.004 (SE 0.007), directionally consistent with sorting but statistically insignificant. Only 8 municipalities adopted a tax during the panel with sufficient LODES employment data, providing insufficient power for the extensive margin. The effect size is small relative to the intensive-margin results in Q2, though the comparison is imprecise. We turn to rate *changes* among already-taxed municipalities, where sample sizes are substantially larger.

5.2 Q2: Do Rate Increases Cause Sorting?

Table 3: Q2: Effect of Rate Increases on share_live_in

	Inc vs No-Tax		Inc vs Stable-Rate	
	(1) OH	(2) PA	(3) OH	(4) PA
Sun-Abraham (binary treatment)				
ATT	-0.0074^{**}	$-0.0091^{\cdot}$	-0.0001	$-0.0053^{\cdot}$
	(0.0025)	(0.0046)	(0.0021)	(0.0029)
DCDH (continuous treatment)				
Avg Total Effect	-0.2314	—	0.0446	-0.0078
	(0.3863)		(0.3132)	(0.0069)
Placebo p	0.406	—	0.147	0.904
Treated	252	157	252	196
Control	104 no-tax	5 no-tax	403 stable	2,209 stable

Notes: Columns organized by control group. "No-tax" = municipalities that never levied a local income tax. "Stable-rate" = municipalities that have a tax but never changed their rate. Sun-Abraham rows: binary treatment (1 after first rate increase). DCDH rows: continuous treatment (actual tax rate) via de Chaisemartin and D'Haultfoeuille (2024) with `continuous(1)`. PA DCDH vs no-tax omitted (only 5 controls). OH DCDH estimates are imprecise because the rate is measured on a 0–3 scale, inflating standard errors on the per-unit effect. PA DCDH vs stable-rate has clean pre-trends (placebo $p = 0.90$). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

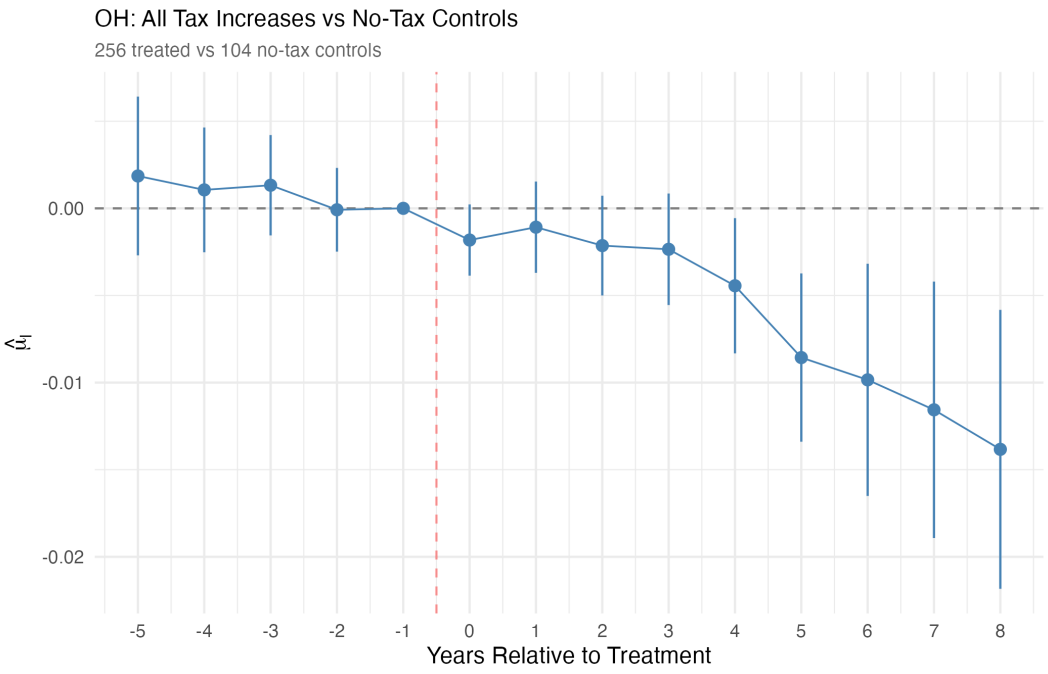


Figure 6: Q2 col (1): OH Rate Increases vs No-Tax Controls. Clean pre-trends; gradual decline through t+8.

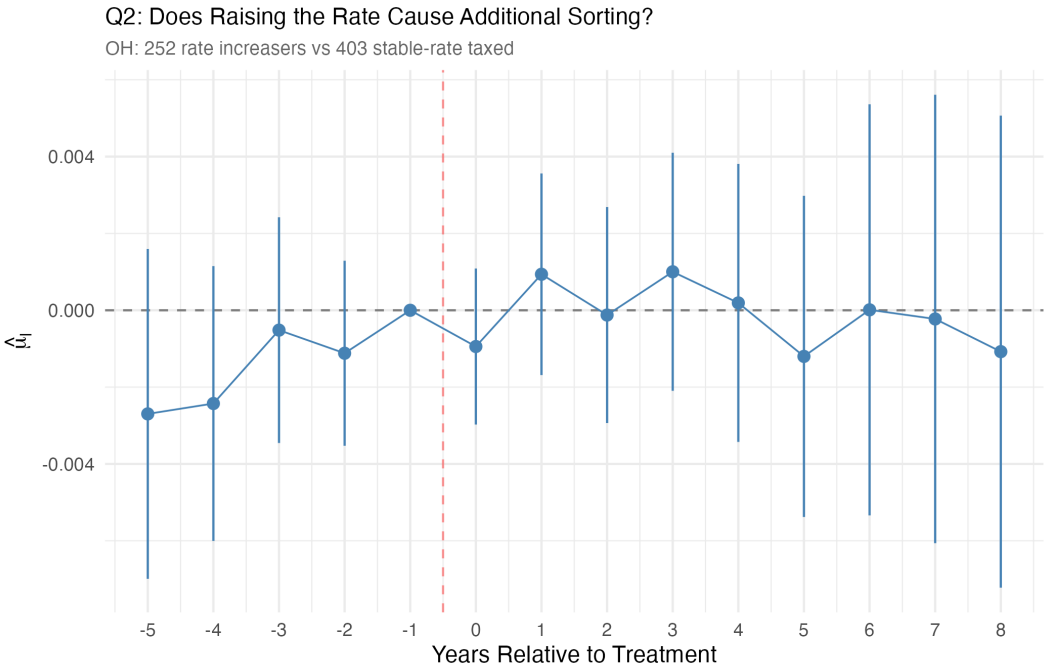


Figure 7: Q2 col (2): OH Rate Increases vs Stable-Rate Taxed Cities. Null — stable-rate cities are poor controls.

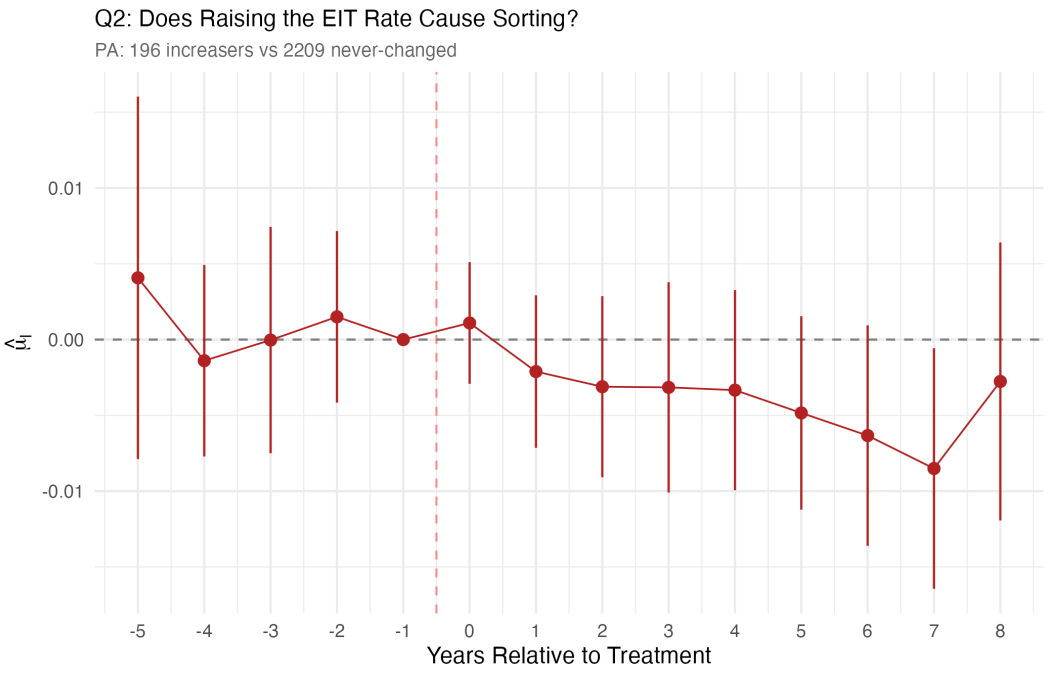


Figure 8: Q2 col (3): PA Rate Increases vs Stable-Rate Municipalities.

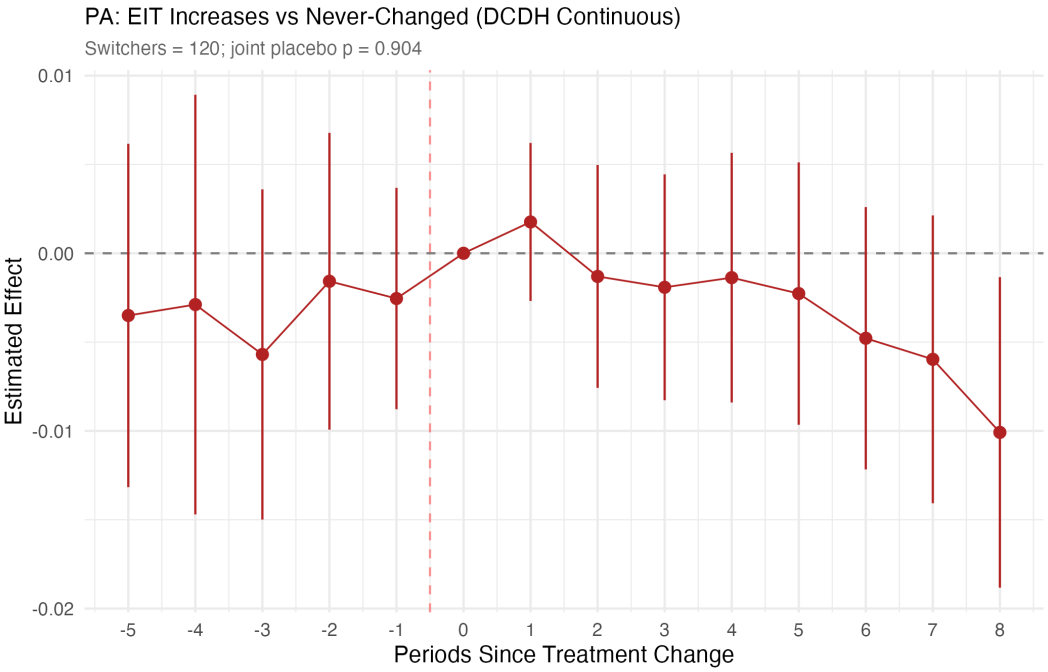


Figure 9: Q2 col (4): PA DCDH Continuous EIT Rate. Clean pre-trends (placebo $p = 0.90$); effect reaches -0.011^{***} at $\ell = 8$.

A2: Yes. Rate increases reduce the locally-resident worker share. The effect is −0.005 to −0.008 per percentage point and builds gradually over 7–8 years. The result is robust across estimators (Sun-Abraham binary and DCDH continuous) and both states. The choice of control group matters for Ohio: no-tax townships produce a significant effect (-0.007^{**}) while stable-rate taxed cities produce a null, suggesting that the latter share confounding urban characteristics with treated cities.

5.3 Q3: Do Rate Decreases Reverse Sorting?

Table 4: Q3: Effect of Rate Decreases on share_live_in

	(1) PA: Dec vs Stable-Rate	(2) PA: Dec DCDH	(3) OH: Dec vs No-Tax
Sun-Abraham ATT	0.0048		−0.0042
	(0.0057)		(0.0064)
DCDH Avg Total Effect		−0.0077	
		(0.0090)	
Treated / Switchers	94	60	36
Control	2,209 NC	NC	104 no-tax
Placebo p	—	0.408	—

Notes: Column (1): PA Sun-Abraham ATT for decreasers vs stable-rate municipalities (null). Column (2): PA DCDH continuous among decreasers (negative ATE — decreases do not increase share_live_in). Column (3): OH decreasers vs no-tax townships (null, 36 treated). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

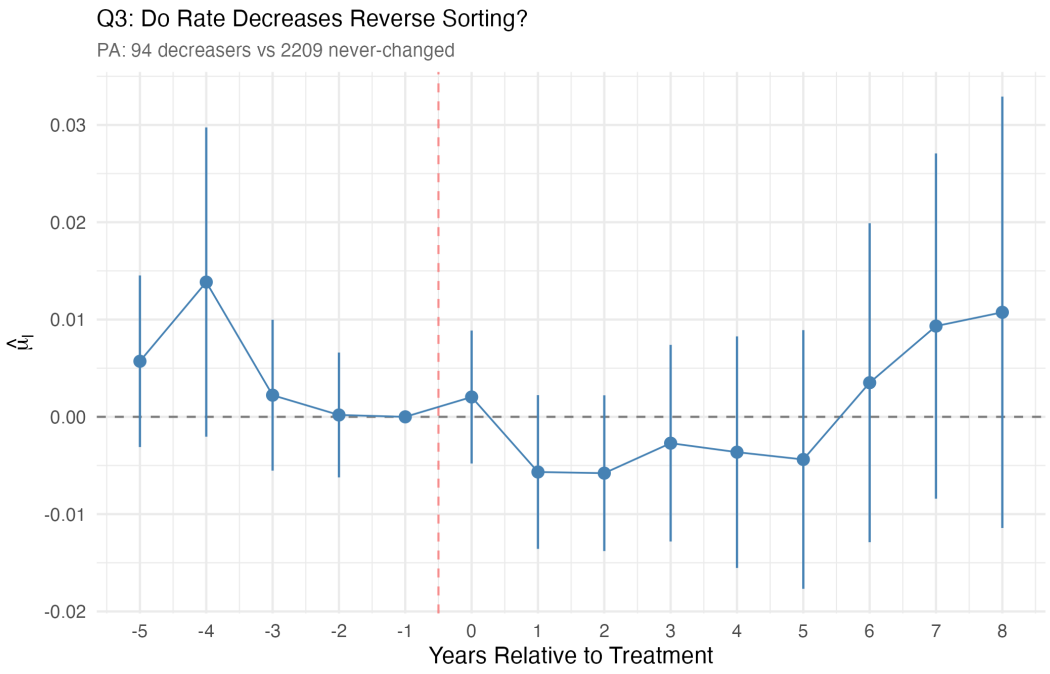


Figure 10: Q3 col (1): PA Rate Decreases vs Stable-Rate. No evidence of reversal.

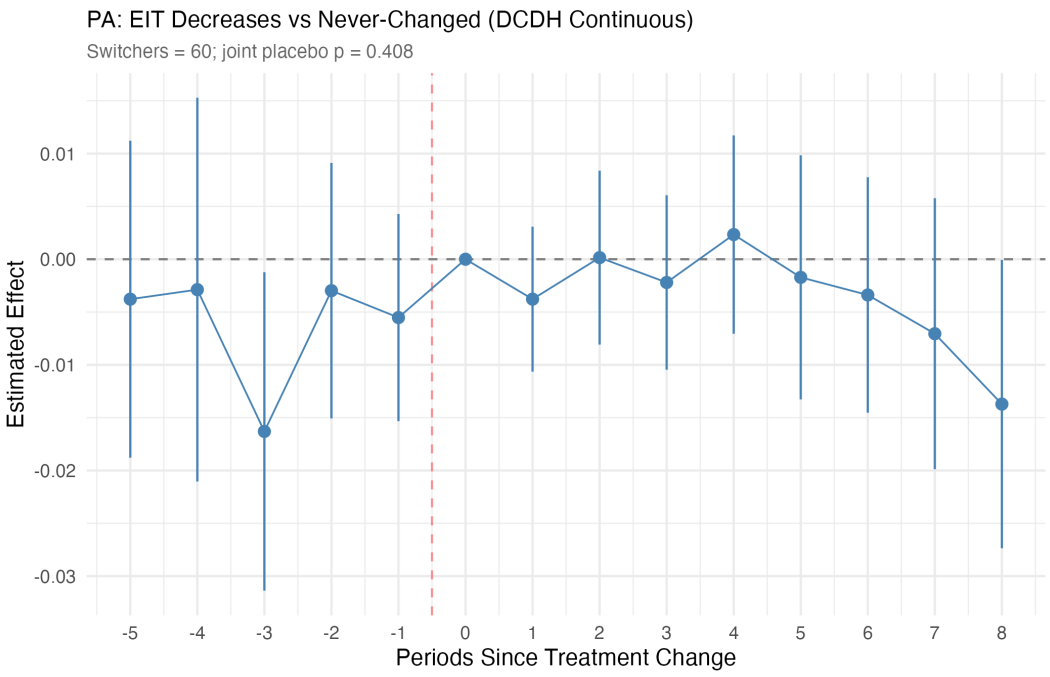


Figure 11: Q3 col (2): PA DCDH Continuous, Decreasers vs Stable-Rate.

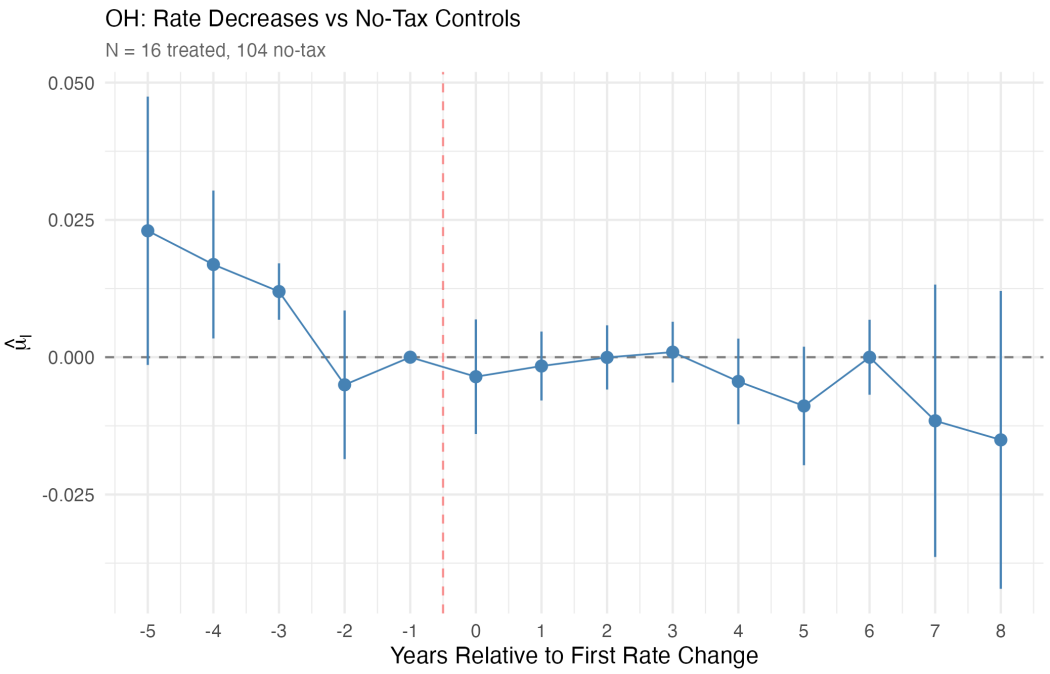


Figure 12: Q3 col (3): OH Rate Decreases vs No-Tax. Wide confidence intervals.

A3: No. Rate decreases do not reverse sorting. All decrease estimates are statistically indistinguishable from zero in both states. The PA DCDH estimate is actually negative (-0.008), suggesting that municipalities that cut their rates continue to lose locally-resident workers, possibly because rate cuts signal fiscal distress.

5.4 Q4: Is the Response Asymmetric?

Table 5: Q4: Asymmetry Test — Joint Increases and Decreases

	Pennsylvania		Ohio	
	(1)	(2)	(3)	(4)
	Joint Model	Split Dose	Joint Model	Split Dose
inc × post	-0.0074^{*}		-0.0055^{**}	
	(0.0037)		(0.0020)	
dec × post	−0.0002		−0.0080	
	(0.0060)		(0.0126)	
R^{+} (cum. inc)		-0.0177^{*}		−0.109
		(0.0069)		(0.379)
R^{-} (cum. dec)		0.0012		−1.155
		(0.0040)		(1.353)
N	24,526	24,526	5,081	7,718

Notes: Columns (1), (3): joint model with increase and decrease post indicators. Columns (2), (4): split cumulative dose on full sample. PA: rate increases reduce share_live_in ($\beta^{+} = -0.018^{*}$ per pp); rate decreases have no effect ($\beta^{-} \approx 0$). OH split dose is imprecise (0–3 rate scale). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

A4: Yes, the response is asymmetric. Rate increases reduce the locally-resident share ($\delta^{+} = -0.007^{*}$ in PA, -0.006^{**} in OH; $\beta^{+} = -0.018^{*}$ per pp in the split dose), but rate decreases have no effect ($\delta^{-} \approx 0$, $\beta^{-} \approx 0$). This pattern is consistent across both states, both estimators (joint model and split dose), and the DCDH results from Q3. Residential location decisions are sticky: once a tax increase pushes workers to relocate their residence outside the taxing municipality, cutting the rate does not bring them back.

5.5 Companion Outcomes: Decomposition

Table 6: Companion Outcomes: Decomposing the share_live_in Effect

	(1)	(2)	(3)	(4)
	share_live_in	log(Resident)	log(Nonresident)	log(Total Jobs)
PA Dose-Response (treated municipalities only)				
rate × post	−0.0099·	−0.1530*	−0.1012·	−0.1172*
	(0.0052)	(0.0720)	(0.0548)	(0.0549)
post	0.0022	0.1226·	0.1113·	0.1173*
	(0.0064)	(0.0721)	(0.0596)	(0.0587)
Diagnostics				
N	2,741	2,741	2,741	2,741
R^2	0.912	0.979	0.980	0.982
Within R^2	0.011	0.015	0.008	0.011
Fixed Effects		Municipality, Year		
Clusters		Municipality		
Equality test (2) vs (3)		$p = 0.09$		

Notes: Dose-response specification estimated among PA municipalities that raised EIT rates, 2002–2019. Column (1): main outcome. Columns (2)–(4): log counts of resident workers (live and work in municipality), nonresident commuters (work in municipality, live elsewhere), and total jobs. Equality test: joint model testing whether the resident coefficient differs from the nonresident coefficient. $\cdot p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

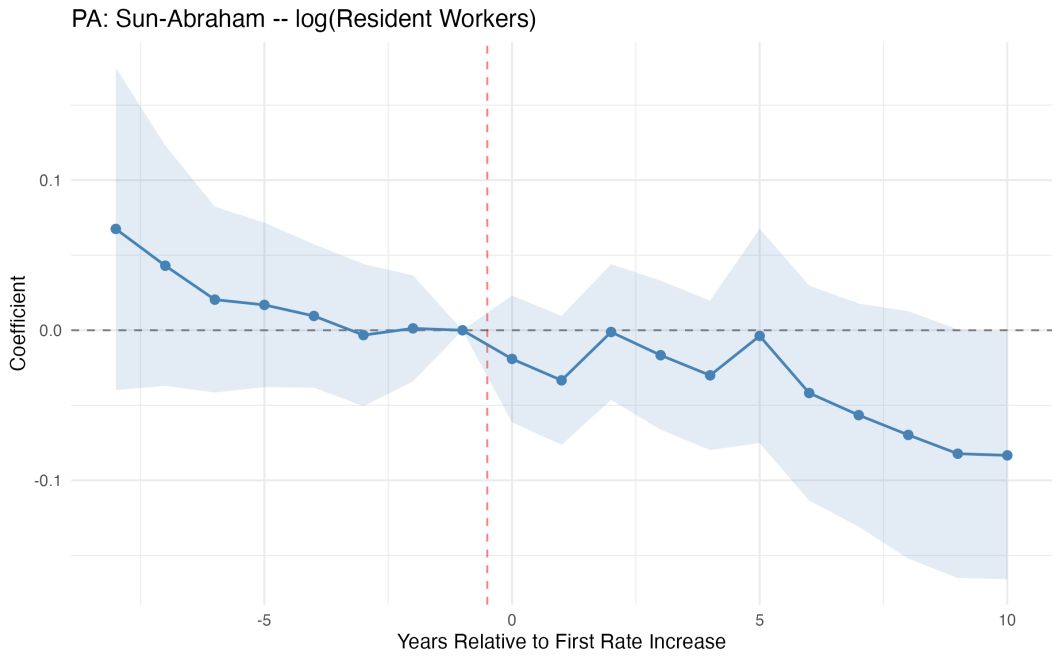


Figure 13: Pennsylvania: Sun-Abraham Event Study — log(Resident Workers).

5.6 Fiscal Impact

Table 7: Fiscal Impact of EIT Rate Changes (PA, 2002–2019)

	(1) log(EIT Rev)	(2) log(Total Tax)	(3) log(Total Rev)	(4) log(Police)	(5) log(Pop)	(6) log(Mkt Val)
resident_eit	0.6710**	0.2957**	0.0643	0.0531	−0.0094	0.0066
	(0.2106)	(0.0948)	(0.0429)	(0.0461)	(0.0137)	(0.0218)
N	42,309	42,526	42,543	22,979	42,523	9,323
R^2	0.977	0.983	0.959	0.953	0.997	0.978
Within R^2	0.027	0.006	0.0001	0.0001	0.0001	0.000
Fixed Effects	Municipality, Year					
Clusters	Municipality					

Notes: PA DCED Annual Financial Reports, 2002–2019. All outcomes in logs. Column (6) limited to 2015–2019 (market value data availability). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

5.7 Heterogeneity

6 Robustness

Table 8: Robustness: Fiscal Controls and Philadelphia Exclusion

	TWFE			Dose-Response (treated only)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	+ Controls	No Philly	Baseline	+ Controls	No Philly
resident_eit	-0.0126^{*}	-0.0104^{*}	-0.0165^{**}			
	(0.0052)	(0.0047)	(0.0060)			
rate × post				-0.0089^{**}	-0.0079^{**}	-0.0098^{**}
				(0.0030)	(0.0028)	(0.0032)
N	27,277	14,174	27,257	3,058	2,173	3,038
Controls	No	Yes	No	No	Yes	No
Philadelphia	Yes	Yes	No	Yes	Yes	No

Notes: Controls include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid from PA DCED Annual Financial Reports. Philadelphia excluded in columns (3) and (6) due to its structurally different wage tax system. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

7 Discussion

8 Conclusion

Acknowledgments

References

[1] Tiebout, C. M. (1956). A Pure Theory of Local Expenditures. *Journal of Political Economy*, 64(5), 416–424.

[2] Sun, L. & Abraham, S. (2021). Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics*, 225(2), 175–199.

[3] Callaway, B. & Sant'Anna, P. H. C. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2), 200–230.

A Pre-Treatment Balance

Table 9: Pre-Treatment Balance: Treated vs. Never-Treated PA Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	N (T / C)
Population	14,174	3,711	0.139	1,265 / 39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267 / 39,692
EIT Revenue ($)	4.4M	438K	0.062	
RE Tax Revenue ($)	2.9M	538K	0.129	
Police Expenditure ($)	3.1M	424K	0.110	
Streets Expenditure ($)	1.5M	496K	0.121	
EIT Rate (%)	0.50	0.50	0.055	

Notes: Treated = municipalities that raised EIT during the sample period (pre-treatment years only). Control = municipalities that never raised EIT. All normalized differences < 0.15, within standard balance thresholds. Source: PA DCED Annual Financial Reports.

B Pre-Trend Tests on Fiscal Observables

We test the parallel trends assumption on six fiscal observables from the PA DCED Annual Financial Reports using the Sun and Abraham (2021) estimator. For each outcome, we estimate Equation 1 with municipality and year fixed effects, clustering standard errors at the municipality level. Table 10 reports the maximum absolute t-statistic among the pre-treatment coefficients at relative times $\ell \in \{-5, -4, -3, -2\}$.

Pre-trends are flat for five of six fiscal observables: total revenue (max $|t| = 1.60$), EIT revenue (1.33), streets expenditure (0.80), population (1.09), and total debt (1.36). None of the pre-treatment coefficients for these outcomes are individually significant at the 5% level.

Police expenditure is the exception. The pre-treatment coefficients at $\ell = -5$ and $\ell = -4$ are negative and significant ($t = -2.85$ and $t = -2.42$, respectively), indicating that treated municipalities had relatively lower police spending 4–5 years before the rate increase. This pattern suggests that some EIT increases may have been associated with anticipated public-safety expansions financed by the new revenue. Importantly, the main results on `share_live_in` are robust to controlling for log police expenditure (Table 8), confirming that this pre-trend does not confound the estimated sorting response.

Table 10: Pre-Trend Tests: Sun-Abraham on Fiscal Observables (PA, 2002–2019)

	Pre-Treatment Coefficients					
Outcome	$\ell = -5$	$\ell = -4$	$\ell = -3$	$\ell = -2$	Max $\lvert t \rvert$	N
log(Total Revenue)	−0.076	−0.051	−0.012	−0.052	1.60	43,170
	(0.049)	(0.056)	(0.038)	(0.032)		
log(EIT Revenue)	−0.025	−0.033	0.027	0.049	1.33	42,278
	(0.051)	(0.052)	(0.032)	(0.037)		
log(Police Exp)	-0.085^{***}	-0.068^{**}	0.011	0.037	2.85	23,464
	(0.030)	(0.028)	(0.052)	(0.039)		
log(Streets Exp)	−0.001	0.003	−0.043	−0.029	0.80	43,105
	(0.059)	(0.075)	(0.053)	(0.059)		
log(Population)	0.011	0.007	0.002	0.003	1.09	43,150
	(0.012)	(0.008)	(0.006)	(0.003)		
log(Total Debt)	−0.029	−0.051	−0.179	−0.103	1.36	24,863
	(0.176)	(0.214)	(0.132)	(0.102)		

Notes: Each row reports Sun and Abraham (2021) estimates of the pre-treatment coefficients $\hat{\mu}_\ell$ for $\ell \in \{-5, -4, -3, -2\}$, with $\ell = -1$ as the reference period. Standard errors (in parentheses) are clustered at the municipality level. Max $|t|$ is the largest absolute t-statistic among the four pre-treatment coefficients. Pre-trends are flat for all outcomes except police expenditure, where treated municipalities had significantly lower spending at $\ell = -5$ and $\ell = -4$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PA DCED Annual Financial Reports.

C Additional Figures